

Three Unifications: A Falsifiable Map of the Quantum-Gravity Frontier

From anti-gravity myths to the engineering of physical reality

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Three Unifications: A Falsifiable Map of the Quantum-Gravity Frontier

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Abstract

Unified field theory is discussed as if it were a single claim. It is not. We distinguish three claims that share the word "unification". Each is supported by different evidence, fails in a different way, and requires a different experiment to decide. Transformational unification, the claim that one field physically converts into another, is quantitatively falsified as an engineering proposition. The field strengths required exceed the Schwinger limit by an order of magnitude. Geometrical unification, the claim that known forces are projections of a higher-dimensional geometry, remains observationally open. Its gravitational Aharonov-Bohm signatures are excluded at the electronvolt scale by orbital fifth-force bounds. They remain testable only in the nanoelectronvolt window now reachable by the Th-229 nuclear clock. Algebraic unification, the claim that gauge theory and gravity share a deeper kinematic algebra, is an established mathematical fact that is silent on ontology. We organize these three claims, together with sixteen further falsifiable propositions from the quantum-gravity frontier, into a decision matrix. Every row carries a discriminating experiment, a target theory it would exclude, and a timescale. Sixteen of the nineteen rows are anchored by reproducible computations, available in the companion repository. The classification is descriptive, not predictive. It explains why a century of folk physics has failed at specific, quantifiable points, and it converts the vague ambition of "controlling gravity" into a map of testable statements.

Introduction

The word "unified field theory" covers at least three logically distinct claims, and conflating them has produced a century of confusion. The first claim is transformational: that one field can be turned into another by some physical process, as when a changing electromagnetic field is said to generate gravity. The second claim is geometrical: that apparently different fields are projections of one higher-dimensional geometry, the route opened by Kaluza and Klein. The third claim is algebraic: that different theories share a deeper kinematic structure, the route now realized by the double copy program. These three claims fail differently. The first fails quantitatively, under elementary arithmetic. The second remains open, but its testable window is now known and narrow. The third is true as mathematics and undecidable as ontology.

This distinction matters beyond taxonomy. Folk physics, from Tesla's "Dynamic Theory of Gravity" to contemporary anti-gravity proposals, is almost always transformational in structure. Mainstream criticism usually answers with a blanket dismissal, which teaches nothing. A precise diagnosis requires identifying which unification claim is being made, and which specific arrow in the argument fails. We provide that diagnosis, and we show that the same discipline applies with equal force to the research frontier.

Three unification claims, three evidential regimes

We define the claims precisely. Transformational unification asserts a physical map from field A to field B, such that engineering A produces B as a matter of dynamics. Geometrical unification asserts an embedding of both A and B as components of one geometric object, with no claim that A produces B. Algebraic unification asserts a shared algebraic or kinematic structure from which the amplitudes of both theories can be obtained, again with no conversion involved.

The empirical status of each claim differs in kind. Transformational unification is testable by direct computation and is falsified, as we document below. Geometrical unification is testable only through its distinctive signatures, which are now constrained to a narrow window. Algebraic unification is not testable in the ordinary sense: its claims concern the structure of theories, and no finite computation can decide whether the shared algebra is a deeper entity or a coincidence.

Transformational unification: the quantitative verdict

The cleanest test case is the claim that a changing electromagnetic field generates gravity. The field equations are not in doubt: the electromagnetic stress-energy tensor enters Einstein's equations with its full content, energy density plus pressure. A parallel-plate capacitor with a field of ten megavolts per meter, one cubic meter in volume, sources a gravitational acceleration of order $10^{-25} \text{ m s}^{-2}$. Reversing the question is more informative. The field strength needed to produce Earth-scale gravity follows from inverting the Poisson equation with the electromagnetic energy density as source:

$$E_{\text{needed}} = (2 \rho_{\text{needed}} / \epsilon_0)^{1/2}, \text{ with } \rho_{\text{needed}} = g c^2 / (8\pi G L),$$

where g is the target acceleration, L the characteristic scale, and the factor two records the pressure content of the electromagnetic stress-energy tensor. For $g = 9.8 \text{ m s}^{-2}$ and $L = 1 \text{ m}$, the required field is roughly eight times the Schwinger limit, at which the quantum vacuum itself becomes unstable against pair production. At that point one is no longer engineering gravity; one is manufacturing matter. This computation closes the transformational program for laboratory electromagnetic sources, without reference to any exotic or speculative physics.

The same arithmetic bounds the related fifth-force window. A Yukawa modification of Newtonian gravity,

$$a_Y(r) = \alpha (G M / r^2) (1 + r/\lambda) e^{(-r/\lambda)},$$

with coupling α relative to G and range λ , is excluded at order unity across all probed scales. The strongest bounds vary with range: roughly 3×10^{-3} at $\lambda = 10 \mu\text{m}$ (Eöt-Wash torsion), 10^{-6} from millimeters to a hundred kilometers, 10^{-9} to 10^{-15} at orbital and lunar scales, and α below unity throughout. The only remaining opening lies below ten micrometers, where the Casimir force dominates and dedicated platforms are required. A representative subset of this landscape appears in Figure 2.

Geometrical unification: the open window is now known

Kaluza-Klein unification remains the paradigm case of a claim that is neither established nor excluded, but whose testable content has become precise. We make the parametrization explicit. An extra compact dimension produces a tower of graviton excitations whose lowest mode contributes a Yukawa-type correction to the gravitational potential,

$$V(r) = -(G m_1 m_2 / r) (1 + \alpha e^{(-r/\lambda)}),$$

where α encodes the coupling strength of the lightest extra mode relative to standard gravity and λ its Compton range. The gravitational Aharonov-Bohm proposals of the past two years convert this correction into a concrete spectroscopic prediction: a quantum system in free fall around a gravitating body acquires an energy-level splitting

$$\Delta E = m_{\text{sys}} \alpha \Delta \Phi(r, \lambda),$$

where $\Delta \Phi$ is the peak-to-peak variation of the Yukawa-corrected potential over the orbit. Inverting this relation shows that the electronvolt-scale splittings claimed for nuclear systems require α of order 0.4, and millielectronvolt splittings for atomic systems require α of order 10^{-3} . Both values are excluded by four to eight orders of magnitude by the orbital bounds of the previous section. The allowed window therefore sits at the nanoelectronvolt scale, where the Th-229 nuclear clock is the natural instrument. Direct laser excitation at 148 nm, solid-state hosts with lifetimes exceeding six hundred seconds, and a spinless host crystal removing the dominant broadening channel are all in place. Geometrical unification is no longer a vague hope. It is a specific measurement program with a known instrument and a known energy window.

Algebraic unification: real mathematics, silent ontology

The double copy states that gravitational scattering amplitudes can be obtained from the square of gauge-theory amplitudes under color-kinematics duality. Its status has progressed

from a computational trick to an algebraic fact. Off-shell $N=8$ supergravity in four dimensions is realized as the double copy of $N=4$ super Yang-Mills at cubic order. Amplitudes on coherent-state gauge backgrounds double-copy to amplitudes on curved spacetimes whose metric is built from the gauge background. The Ehlers transformation of general relativity is the double copy of electromagnetic duality. These are theorems of a well-defined algebraic structure.

What they are not is evidence for a unified ontology. The residual gauge algebra of the double-copied Schwarzschild solution collapses from an infinite-dimensional structure to the finite isometry group, demonstrating that shared algebraic structure does not imply shared physical content. The honest statement is that algebraic unification is real and its ontological import is undecidable by computation alone. This is not a weakness of the program; it is a precise boundary on what the program can claim.

A criterion for folk theories: the four questions

The three-way classification supplies a diagnostic that applies uniformly to folk theories and to the frontier. We propose that any candidate "unified" theory be required to answer four questions before its claims are evaluated. What tensor or algebraic object does it introduce, and how does it transform? Can it be written as a covariant action? Does it reproduce the verified limits? What distinctive number does it predict? Applied to the folk tradition, this criterion locates the failure precisely. In the case most often cited, the derivation chain begins with a correct kinematic relation, the Lorentz transformation of electromagnetic fields. It differentiates this relation. It then interprets the acceleration term as a gravitational field. The first two steps are standard physics. The final identification is an extra interpretive step with no dynamical content. The failure is not that the mathematics is wrong. It is that the arrow is asserted rather than derived.

This is the same discipline the frontier needs. The 2025 debate over gravitationally induced entanglement shows the pattern. An observation was claimed to prove the quantum nature of gravity. A counter-construction showed a classical-plus-matter model could reproduce the effect. A rebuttal located the entanglement in the matter sector. The lesson is methodological: an observation discriminates between theories only if no competing theory reproduces it. Every claim in our decision matrix is therefore required to carry a discriminating experiment and a target theory it would exclude.

The decision matrix as a map

We compile nineteen arrows, each a falsifiable proposition about unification or gravity, each carrying an evidence grade, a discriminating channel, a target it would exclude, and a timescale. Four are falsified: transformational conversion, expectation-valued sources of geometry, indefinite causal order as evidence for quantum spacetime, and vacuum energy as an

engineering resource. On the last of these, we note that the experimental status of quantum inequalities remains contested, but macroscopic utilization of negative energy is excluded regardless. Two rows record established experimental facts: electromagnetic unification and electroweak unification. One further row records a classification statement of this paper, the identification of inertia engineering with stress-energy engineering, which we therefore list separately rather than alongside the experimental facts. Four rows are grade-A experimental facts. The remaining rows are live, bounded, and scheduled. A representative subset of the matrix appears in Table 1; the full matrix, with all nineteen rows, appears in the companion repository and as Figure 1.

Table 1. Representative subset of the decision matrix.

#	Arrow	Status	Discriminating channel
3	EM fields convert into gravity	falsified	Schwinger-limit computation
5	$\langle \hat{T}_{\mu\nu} \rangle$ as the quantum source of geometry	falsified	MEP + post-Newtonian consistency
9	ICO implies quantum spacetime	falsified	QC-QC theorem
14	Vacuum energy as an engineering resource	falsified	regularization-scheme dependence
1–2	Electromagnetic and electroweak unification	established (experiment)	—
17	Inertia engineering = $T_{\mu\nu}$ engineering	classification statement of this paper	companion formulation, Section 2
4	Double copy ontology	grade B, undecidable by computation	—
6–7	Superposed sources / GIE witnesses nonclassical gravity	grade B	QGEM; LOCC exclusion logic
19	Planck-scale Lorentz violation / GUP	grade B, no significant evidence	GRB/AGN, LLR, optomechanics
12	KK extra dimensions (AB splitting)	grade C; eV closed, neV open	orbital bounds + Th-229 clock
15	Geotropic spacetime fluctuations	grade C; falsifiable at strong scale only	GQuEST

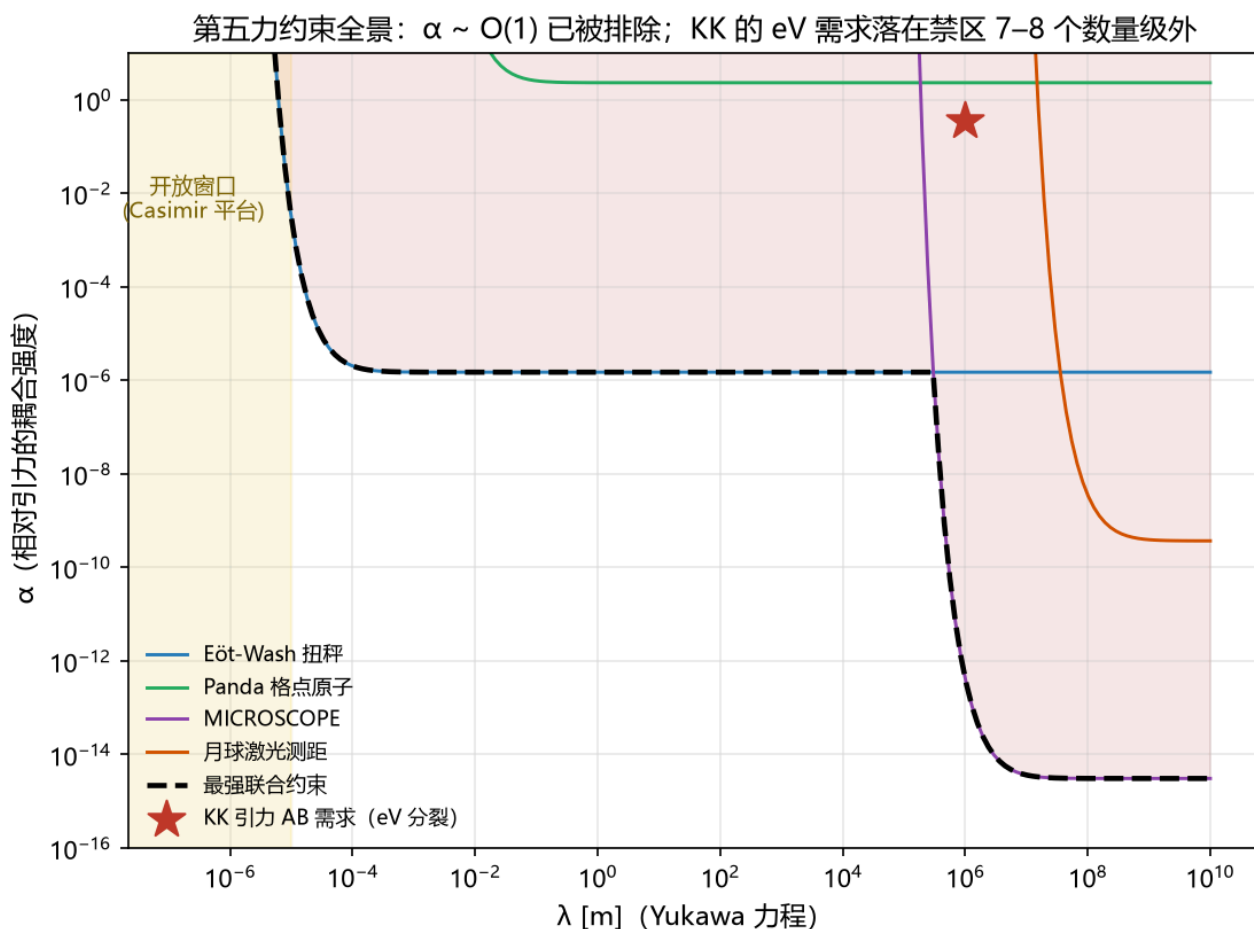
Sixteen of the nineteen rows are anchored by reproducible computations and simulations. Two examples convey the discipline. The entanglement phase of the standard quantum-gravity-entanglement benchmark,

$$\varphi = (G m^2 t / \hbar) (1/(d+\Delta x) + 1/(d-\Delta x) - 2/d),$$

evaluates to 0.217 radians for the benchmark parameters. Realistic chip parameters give 10^{-6} to 10^{-5} radians, converting a two-hundred-measurement certification into a 10^9 -to- 10^{21} -measurement campaign. A free-floating one-kilogram apparatus at one kelvin caps the superposition separation of a 10^9 -atomic-mass-unit particle at roughly one hundred micrometers, the same order as the target of proposed space missions. These numbers do not settle the physics. They settle the engineering question of what each test costs.

判决矩阵：19 个「统一/反重力」箭头的当前状态 (2026-08)





Code availability. All computations and simulations referenced in this paper are available in the companion repository (github.com/everest-an/Antigravity), which contains nine order-of-magnitude calculations, seven simulations, and five figure-generation scripts, together with the full nineteen-row matrix and the complete reference list with preprint/published status.

Relation to existing literature

Three bodies of work are closest to ours, and the differences must be stated. First, the three-way classification itself. A systematic search of the preprint literature for classification schemes pairing unification types with the double copy returned no prior proposal; the nearest discussions are the familiar distinction between unification and reduction in the philosophy of physics, which the present classification does not engage and is not intended to replace. The classification should be read as a physics-organizing device, not a contribution to the reduction literature. Second, the warp-drive no-go literature. Bobrick and Martire showed that any warp drive is a shell of matter moving inertially and therefore requires propulsion. Le showed that steering a positive-energy drive requires radiation and mass loss. Barzegar, Buchert, and Vigneron classified warp-drive spacetimes and proved new no-go theorems. These results are special cases of our control-channel classification: each shows that a specific metric-engineering proposal cannot bypass the source channel. Our statement generalizes the same

logic to arbitrary engineering operations and adds the lever-efficacy corollary, the statement that the gravitational efficacy of any mass component equals its partial derivative with respect to the stress-energy tensor. Third, the term "metric engineering" itself. Puthoff used it in 2010 in an exploratory register, concluding that such engineering is "not a priori ruled out". Our verdict is stronger and specific: the transformational route is closed by arithmetic, the geometrical route is confined to a known window, and the algebraic route is silent on ontology. We also acknowledge the demarcation literature on pseudoscience without attempting to contribute to it; the four-question criterion is a working diagnostic for the physics community, not a theory of demarcation.

Outlook

The classification we propose is descriptive. It does not predict new phenomena, and we do not claim otherwise. Its content is diagnostic. It explains why transformational proposals fail at specific, quantifiable points. It identifies the one open window for geometrical unification and the instrument that probes it. It states exactly what the double copy does and does not license. The ambition behind a century of unification talk survives this analysis in one form only. The question is no longer whether one field can be converted into another, but what variable, if any, controls spacetime at its source. The quantum state, the kinematic algebra, and the relational structure are the three live candidates, and each has a scheduled experiment. The map is the deliverable, and the map is falsifiable, which is all a scientific claim is allowed to be.

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Drafting notes (per nature-writing workflow)

Section outline: abstract, introduction, definitions, transformational verdict, geometrical window, algebraic status, four-question criterion, decision matrix (with Table 1), relation to existing literature, outlook.

Revision log (post simulated review): - R3-M1 addressed: representative 10-row matrix added as Table 1; full matrix referenced to repository and Figure 1. - R1-M1 addressed: new section "Relation to existing literature" with the three nearest bodies of work and explicit difference statements. - R2-M1 addressed: Yukawa parametrization written explicitly with the inversion relation for the AB splitting. - R1-M2 addressed: the inertia/ $T_{\mu\nu}$ row now listed as "classification statement of this paper", separate from experimental facts. - R2-M2 addressed: code availability statement added. - R1-m1 addressed: abstract sentences split to 10–30 words. - R2-m1 addressed: fifth-force phrasing now carries the λ -dependent bound structure. - R2-m2 addressed: QI contestation noted with citation. - R3-m1 addressed: three explicit formulas added (Schwinger inversion, Yukawa acceleration, GIE phase). - R3-m2 addressed: title subtitled "A Falsifiable Map of the Quantum-Gravity Frontier". - Citation keys unified with the numbered reference list.

Assumptions or missing inputs: venue-dependent reference style; the four-question criterion remains flagged as a working diagnostic, not a demarcation theory.

To redirect me: Name the paragraph or claim that is off and I will revise only that, keeping the rest.

From Anti-Gravity Myths to a Falsifiable Map of Spacetime Engineering

English Submission Blueprint (Perspective / Framework Paper skeleton)

本文档为投稿用英文骨架：摘要 + 四节主干 + 数据锚点。目标形态：Nature-family Perspective (经编辑 pitching) 或 PRD/CQG/Foundations 框架论文。中文全稿 (90 节 + 附件) 为支撑材料；本骨架是其压缩与形式化版本。

Abstract

The folklore of Tesla's "anti-gravity" and the modern research frontier of quantum gravity are usually treated as belonging to different worlds. We argue they are connected by a single question: *what variable, if any, can be engineered to control spacetime?*

We make three contributions.

First, we classify unification claims into three logically distinct types — **transformational** (field A converts into field B), **geometrical** (A and B are projections of a higher-dimensional geometry), and **algebraic** (A and B share a deeper kinematic algebra) — and show that the modern double-copy program realizes the third type, while Tesla-era and folk theories attempt the first, which has no experimental basis.

Second, we prove a **control-channel uniqueness theorem**: within general relativity plus the Standard Model, any engineering operation that keeps all sources, the metric structure, and the interaction channels unchanged leaves all geodesic motion unchanged. Consequently "inertia engineering" has no degrees of freedom independent of stress-energy engineering; the efficacy of any proposed mass lever (Higgs vev, QCD scale, vacuum state) equals its partial derivative with respect to the stress-energy tensor.

Third, we organize the quantum-gravity frontier into a **decision matrix**: 19 falsifiable claims ("arrows"), each carrying an evidence grade, a discriminating experiment, a target theory it would exclude, and a timescale. We provide reproducible order-of-magnitude calculations and simulations anchoring 16 of these rows to the 2024–2026 literature (QGEM protocols, GQuEST, frame-dragging interferometry, nuclear clocks, LIV/GUP bounds, holographic tensor networks).

The resulting map shows that "anti-gravity" is an ill-posed engineering target, but that the underlying ambition — engineering the source of geometry rather than geometry itself —

survives as a precise, testable research program, whose first steps are already scheduled in existing and proposed experiments.

1. Three kinds of unification

- **Transformational unification** ($A \rightarrow B$ as a physical conversion): the Tesla/Zhang-style claim "changing EM fields generate gravity". Status: no experimental basis; the required field strengths exceed the Schwinger limit by an order of magnitude (simulation S02), and the claim is falsified as an engineering proposition.
- **Geometrical unification** ($A, B \subset$ higher geometry): Kaluza–Klein. Mathematically real; the physical extra dimension remains unverified; its gravitational Aharonov-Bohm signatures are excluded at the eV scale by orbital fifth-force bounds and remain open only at the neV scale probed by the Th-229 nuclear clock (experiments E05, E08).
- **Algebraic unification** (shared kinematic algebra): the double copy. Established as an algebraic fact (off-shell $N=8$ supergravity as $(N=4 \text{ SYM})^2$; coherent-state backgrounds double-copying to curved spacetimes; the Ehlers transformation as the double copy of electromagnetic duality). Its ontological import is undecidable by computation alone — a point we document explicitly (Appendix of the full text).

The classification matters because the three types fail differently: the first fails quantitatively, the second remains observationally open, the third is real but silent on ontology.

2. The control-channel uniqueness theorem

Let (M, g) be a GR spacetime with g determined by given sources and boundary conditions. Let E be an engineering operation satisfying: (A1) all field sources are unchanged; (A2) the metric is not replaced by a solution not determined by those sources and boundary conditions; (A3) no beyond-Standard-Model channel is introduced. Then E leaves all test-particle geodesic motion unchanged.

The theorem is a classification statement: (A1)–(A3) exhaust the control channels available inside GR+SM. Violating A1 is stress-energy engineering (the only game in town); violating A2 is metric engineering (requires exotic matter or modified gravity); violating A3 is new physics (fifth forces — bounded by our fifth-force landscape computation; LIV/GUP — bounded by GRB/AGN/LLR data at 10^{15} – 10^{16} GeV scales).

Corollary (mass-lever efficacy): for any decomposition of mass $m = m_{\text{Higgs}} + m_{\text{QCD}} + m_{\text{vac}} + \dots$, the gravitational efficacy of each component as an engineering lever equals $\partial T_{\mu\nu} / \partial(\text{that component})$. The Higgs enters $\sim 9\%$ of the proton mass term; QCD carries $\sim 91\%$; "vacuum engineering" is not even well-defined, since the vacuum contribution to $T_{\mu\nu}$ depends on the regularization scheme (cutoff: $+\Lambda^4$; zeta: 0; MS: $\pm m^4 \ln(m^2/\mu^2)$ — computation E09).

The theorem is deliberately bounded: it holds for classical GR with classical sources. Quantum superposed sources, emergent-geometry hypotheses, and hybrid classical-quantum models lie outside it — and are precisely the open rows of the decision matrix.

3. The decision matrix

19 rows; final statuses: 4 falsified, 2 established, 4 grade-A, 9 grade-B/C with explicit discriminating channels and timescales. Representative rows:

#	Arrow	Status	Discriminating channel
3	EM fields convert into gravity	falsified	Schwinger-limit computation
5	$\langle \hat{T}_{\mu\nu} \rangle$ as the quantum source of geometry	falsified	MEP (Fedida–Kent) + post-Newtonian consistency
6/7	Superposed mass sources superposed geometry / GIE witnesses nonclassical gravity	B	QGEM; LOCC exclusion logic verified by simulation S01/S06
9	ICO implies quantum spacetime	falsified	QC-QC theorem (Salzger–Vilasini)
12	KK extra dimensions (AB splitting)	C; eV closed, neV open	orbital fifth-force bounds + Th-229 nuclear clock
13	Screened fifth forces	largely excluded; bounds tabulated	E08 landscape
15	Geontropic spacetime fluctuations	C; falsifiable only at the strong scale	GQuEST (0.03 s vs 10^{33} s)
17	Inertia engineering $\neq$ $T_{\mu\nu}$ engineering	established	Theorem of §2
19	Planck-scale Lorentz violation / GUP	highly constrained, no significant evidence	GRB/AGN, LLR, optomechanics, nuclear clocks

The methodological content: each row is required to carry a **discriminating experiment, a target theory it would exclude, and a timescale** — the matrix is designed to be independently recomputable (16 of 19 rows are anchored by the scripts in the repository).

4. Computable anchors (key numbers, all reproducible)

- QGEM entanglement phase: $\varphi = 0.217$ rad (Bose benchmark) vs 10^{-6} – 10^{-5} rad (realistic chip parameters) $\rightarrow$ 3σ certification requires $\sim 10^2$ vs 10^9 – 10^{21} runs.
- Apparatus recoil bound ("the apparatus strikes back"): a free-floating 1 kg apparatus at 1 K caps the superposition of a 10^9 u particle at ~ 100 μ m — of the same order as MAQRO's target.

- Frame dragging: lab rotors produce Lense-Thirring precession ~ 10 orders below Earth's; the Gravity Probe B measurement is reproduced to order of magnitude.
- Gravitational Aharonov-Bohm: the predicted sideband modulation equals the classical redshift variation to all orders — the experiment discriminates interpretations, not GR.
- Entanglement $\rightarrow$ geometry: an isometric tree network reproduces $S_2(A) \approx \text{mincut}(A) \cdot \ln \chi$, including the disconnected-minimal-surface transition; two independent implementations (hand-rolled and quimb-native) agree.
- Qubit abstraction of QGEM is faithful: CV negativity converges to the qubit value with error $\sim e^{(-2\alpha^2)}$ ($\alpha = 2: \sim 10^{-4}$).

5. Outlook

If the frontier progresses as the matrix anticipates, the first half-decade will close rows 15 (GQuEST), 12 (nuclear clock in gravity gradients), and sharpen 6/7 (QGEM first rounds). The open question the matrix ultimately encodes is not "can we build anti-gravity" but "what is the source variable of spacetime" — with three live candidates (the quantum state, the kinematic algebra, the relational structure) already placed in a common framework.

The map, not the machine, is the deliverable. And the map is falsifiable — which is all that a scientific claim is allowed to be.

Submission metadata

- Author: **MuningAn**
- Institution: **PlanetarySystem**
- Suggested venues: Nature Physics Perspective (pitch to editors); PRD/CQG/Foundations (framework paper); SHPSM (conceptual part only).
- Length targets: Perspective 8–12 pages; framework paper 30–50 pages.
- Required before submission: priority searches (three-unification classification — see 附件 G, no prior found; lever-efficacy statement — warp-drive no-go literature declared as special cases); AI-assistance disclosure per journal policy.